

Counterfactual Augmented Causal Reasoning for Aspect-Based Sentiment Analysis

Yiming Wu
2410937@tongji.edu.cn
Tongji University
Shanghai, China

Liang Hu*
lianghu@tongji.edu.cn
Department of Computer Science and
Technology, Tongji University
Shanghai, China

Mingzhu Zhou
1120240414@mail.nankai.edu.cn
Nankai University
Tianjin, China

Tangwei Ye
yetw@tongji.edu.cn
Tongji University
Shanghai, China

Xuejie Yang
2512112@tongji.edu.cn
Tongji University
Shanghai, China

Xun Yang
2410914@tongji.edu.cn
Tongji University
Shanghai, China

Zhongyuan Lai
zhongyuan.lai@ballsnow.com
Shanghai Ballsnow Intelligent
Technology Co. Ltd
Shanghai, China

Qi Zhang
zhangqi_cs@tongji.edu.cn
Tongji University
Shanghai, China

Usman Naseem
usman.naseem@mq.edu.au
Macquarie University
Sydney, Australia

Abstract

Aspect-based sentiment analysis (ABSA) aims to identify the sentiment polarity of the specified aspect in a sentence. Despite achieving the remarkable success, deep models are susceptible to capturing spurious correlations between surface patterns and predicted labels instead of capturing aspect-related causal features, leading to poor robustness. To this end, we propose a new counterfactual-enhanced causal reasoning (CECR) framework to reduce spurious correlations. Specifically, at the data level, the counterfactual data augmentation (CDA) module is exploited to employ a three-stage reasoning paradigm guided by cognitive chain-of-thought (COT) principles to generate high-quality counterfactual data from the training data. At the model level, the multi-level causal attention (MCA) is designed to iteratively extract causal features and remove confounding features. Experimental results on the utilized datasets demonstrate that the CECR achieves outstanding performance on state-of-the-art baselines.

CCS Concepts

• Computing methodologies → Causal reasoning and diagnostics.

Keywords

Aspect-based Sentiment Analysis; Large Language Model; Data Augmentation; Causal Inference.

*Corresponding author



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1 Introduction

Aspect-based sentiment analysis (ABSA), as a fine-grained sentiment task, has garnered growing interest in both academia and industry [51]. The aim of the ABSA task is to identify the sentiment polarity of a specified aspect in a given sentence. To achieve precise sentiment prediction in the fine-grained ABSA task, it is vital to capture viewpoint words describing the specified aspect [49].

So far, deep models achieve superior performance in the ABSA task. They automatically and efficiently learn aspect-related sentiment features from context without depending on time-consuming manual computations [18]. One trend is to design diverse attention mechanisms that construct semantic relationships for learning interactive features [45]. Another trend is to employ graph convolutional networks (GCNs) to model syntactic dependencies between words for capturing aspect-related syntactic awareness features [15]. Recently, pre-trained language models (PLMs), such as BERT [7], provide new breakthroughs in the ABSA task by learning extensive linguistic knowledge from large-scale text corpora [36].

Despite the effectiveness of previous researches, the performance of deep models is susceptible to spurious correlations between surface patterns and predicted labels, ignoring the capture of aspect-related causal features [3]. This is because deep models misclassify some contextual semantic features in training data as key features relevant to sentiment prediction when there is a statistical correlation between such features and the predicted labels. For instance,

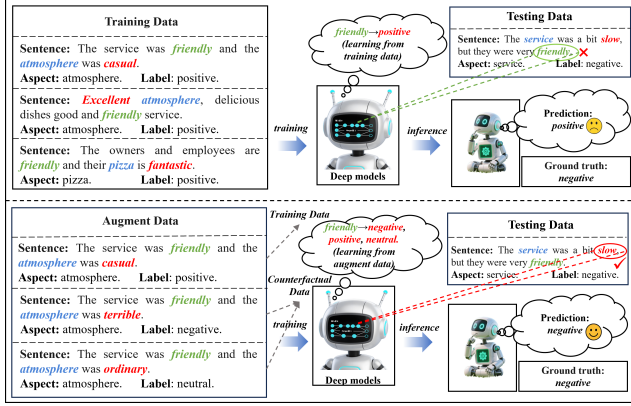


Figure 1: An example of spurious correlation between the contextual word “friendly” and the sentiment polarity “positive” in the restaurant dataset. We use counterfactual data featuring identical spurious context words while different sentiment labels to encourage the model to capture aspect-related causal terms.

as illustrated in Fig. 1, since the contextual word “friendly” in training data is highly co-occurring with the sentiment polarity “positive”, deep models tend to establish strong correlations between “friendly” and “positive” rather than capturing aspect-related causal terms such as “casual”, “excellent”, and “fantastic”. Therefore, the existence of spurious correlations leads to deep models failing to inferring samples in the test data containing the contextual word “friendly” with the ground-truth label “negative”. With such statistical bias, although deep models achieve remarkable performance in IID (identical & independent distribution) situation, they may perform inversely on OOD (out-of-distribution) situation, i.e., poor robustness.

To address these challenges, we propose a counterfactual-enhanced causal reasoning (CECR) framework to capture aspect-related causal features for reducing spurious correlations. At the data level, inspired by the COT reasoning paradigm [50], the counterfactual data augmentation (CDA) module consisting of three steps guides the large language models (LLMs) to execute the identification, modification, and substitution of causal terms, thereby constructing high-quality counterfactual data from the training data. The advantage of this module is to tightly align the logic of counterfactual data generation with human logic, allowing LLMs to build interpretable reasoning processes for making deep models learn counterfactual thinking [36]. For example, as illustrated in Fig. 1, the incorporation of counterfactual data effectively reduces the co-occurrence frequency of the contextual word “friendly” with the sentiment polarity “positive”. Therefore, deep models cannot establish an overly strong correlation between “friendly” and “positive”, encouraging them to capture the causal term “slow” with respect to the specified aspect “atmosphere”. Subsequently, at the model level, the multi-level causal attention (MCA) module consisting of causal semantic retention (CSR) and confounding semantic elimination (CSE) is exploited to capture standard attention for causal features associated with the specified aspect and complementary attention

for confounding features. The two untangled attentions are iteratively optimized in an adversarial minimax fashion, retaining causal features and removing confounding features in an unsupervised fashion to train debiased model. The contributions of this article are as follows:

- We propose a novel CECR framework designed to extract aspect-related causal features to mitigate spurious correlations between surface patterns and predicted labels.
- We propose a CDA module, comprising three sequential steps, designed to guide LLMs in performing the systematic identification, modification, and substitution of causal terms to generate high-quality counterfactual data derived from the original training data.
- We propose a novel MCA module that iteratively extracts causal features while mitigating the influence of confounding features.
- We conduct extensive experiments on five widely utilized benchmark ABSA datasets. Experimental results show that CECR achieves better prediction and robustness performance compared to the strong competitors.

2 Related Work

2.1 Aspect-based Sentiment Analysis

Traditional sentiment analysis, an essential task in natural language processing (NLP), has been oriented to the sentence or document level, which makes difficult to identify the sentiment states of different specified aspects in a sentence or a document [1]. On the contrary, Aspect-based sentiment analysis (ABSA), a fine-grained sentiment analysis task, aims at identifying the sentiment polarity of the specified aspect in a sentence or document. It has achieved remarkable success in domains such as textual [2], audio [28], visual [53], and multimodal [35] scenarios.

The mainstream research trend surrounds the design of different attention mechanisms to model the semantic relationship between the context and the specified aspect for capturing aspect-related sentiment information. Fan et al. [10] proposed a fine-grained and coarse-grained attention framework to achieve word-level interactions between context and the specified aspect, mitigating the effect of information loss caused by long context. Wang et al. [37] proposed an attention-based LSTM to capture the semantic features related to the specified aspect in the context. Chen et al. [4] devised a multiple-attention mechanism to provide tailored memories for different specified aspects to capture distant sentiment features. Tan et al. [29] proposed a dual-attention mechanism with multi-label categorization to recognize complex expressions of opposing viewpoints, addressing the problem of data sparsity. Wang et al. [39] employed a self-attention based variational autoencoder to capture explicit correlations between the specified aspect and contexts.

Another research trend surrounds graph convolutional networks (GCNs) to construct syntactic dependencies between words in a sentence. Wang et al. [37] developed a relational graphical attention network to generate encoded syntactic information by pruning and reconstructing the dependency parsing tree. Singh et al. [25] proposed a dynamic weighting layer to reduce the influence of irrelevant information on sentiment identification and extract

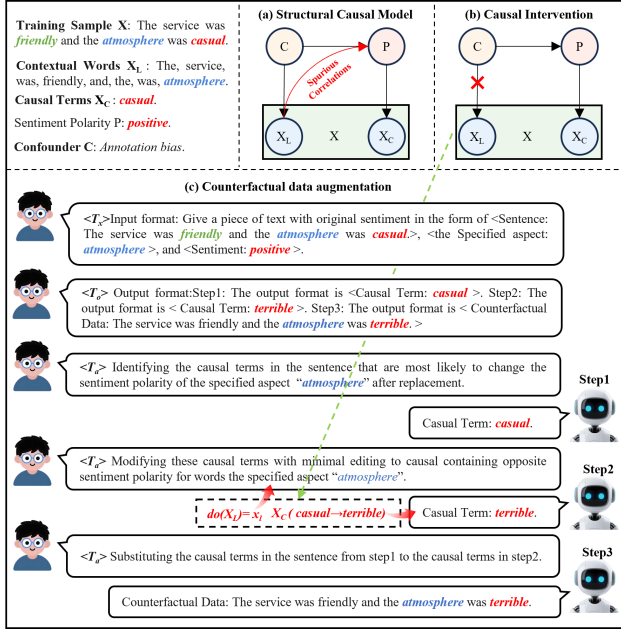


Figure 2: The structural causal model guides counterfactual data generation.

meaningful contextual features. Zhou et al. [54] designed an aspect-centric dependency tree construction approach to minimize the syntactic distance between the specified aspect and its corresponding viewpoints. Yin and Zhong [46] proposed a two-view graphical transformer on text to tightly couple graphical and sequence views to alleviate the oversmoothing problem. However, in real-world scenarios, syntactic dependencies are ineffectively expressed on the informal and complex online reviews. To this end, Li et al. [15] design SemGCN module that can express semantic dependencies to enhance the performance of SynGCN with syntactic dependencies.

2.2 Spurious Correlations in NLP

Despite the preliminary success in NLP tasks, the performance of deep models is susceptible to spurious correlations between surface feature patterns and predicted labels. In fact, due to the presence of specific linguistic patterns in the dependency assumptions [12] or the success of shallow heuristics on most training data [17], deep models achieve promising results without learning underlying generalizations in IID situation. Therefore, deep models tend to have wrong causes but correct results during inference, leading to poor robustness on OOD situation.

Existing approaches to address spurious correlations are mainly categorized into data augmentation [42] and ensemble learning [30]. The core of the data augmentation-based approaches is to construct adversarial samples that surface patterns do not have spurious correlations with the predicted labels, thus training more robust deep models. For example, Zellers et al. [47] proposed adversarial filtering to generate potential counterfactual samples by iteratively training an ensemble of stylistic classifiers to address the challenge of recurring human bias in datasets. Nie et al. [19] devised an NLI

dataset with an iterative and adversarial human-and-model-in-the-loop solution to reveal the vulnerability of deep models for spurious correlations. Wang and Culotta [40] employed statistical matching to identify causal features and generated a debiased dataset using their antonyms for substitution to train a debiased model.

For the ensemble learning-based approaches, the main idea is to train the debiased model by detecting spurious correlations through surface patterns learned by the biased model from the training samples. For example, Stacey et al. [27] proposed joint training of multiple adversarial classifiers ensemble to reduce the performance of all adversarial classifiers, thereby completely removing the assumption bias in sentence representation. Clark et al. [5] proposed an ensemble approach to jointly train low- and high-volume to capture shallow biases and learn generalization patterns. Tian et al. [31] developed a combination of factual and counterfactual models based on causal inference theory to detect spurious correlations in training samples for mitigating data bias.

In this article, the ensemble learning-based approach and data augmentation-based approach are utilized to reduce the impact of spurious correlations for enhancing the robustness of CECR. At the data level, the counterfactual data augmentation (CDA) module is developed to guide LLMs in generating high-quality counterfactual samples for encouraging deep models learn counterfactual thinking. At the model level, the multi-level causal attention (MCA) module is designed to iteratively extract causal features and remove confounding features.

3 Methods

3.1 Task Definition

Given a training data $X = \{(x_i, y_i)\}_{i=1}^N$ containing N samples, each sample x_i comprises the sentence $s = \{w_j^s\}_{j=0}^n$ with n words and the specified aspect $a = \{w_k^s\}_{k=r}^m$ with $m - r + 1$ words, where w_j^s denotes the j th word in the sentence and w_k^s denotes the k th word in the specified aspect. Each sample x_i corresponds to a sentiment label $y_i \in \{y_n, y_o, y_p\}$, where y_n , y_o , and y_p represent negative, neutral, and positive sentiment states. The main purpose of ABSA is to identify the sentiment polarity y_i of the specified aspect a in the training sample x_i .

3.2 Counterfactual Data Augmentation

The structural causal model (SCM) [23, 24, 44] is utilized to construct the causal theoretical foundation for generating counterfactual data.

Structural causal model (SCM). As depicted in Fig.2(a), the SCM is utilized to describe the relationship between causal terms X_C , contextual words X_L , and the sentiment polarity P of the specified aspect. $X_C \rightarrow P$: Causal terms, i.e., viewpoint words describing the specified aspect, are the sole reason for judging the sentiment polarity. $C \rightarrow X_L$ and $C \rightarrow P$: The collection of training data solely involves the aspect-related causal term X_C , ignoring the distribution of contextual words X_L and the sentiment polarity P of the specified aspect. Therefore, annotator bias, i.e., hidden confounders C , may exist to affect the sentiment polarity P of the specified aspect and contextual words X_L .

Spurious correlation. Suppose in the training data, the contextual word x_l “friendly” frequently occurs in sentences that the

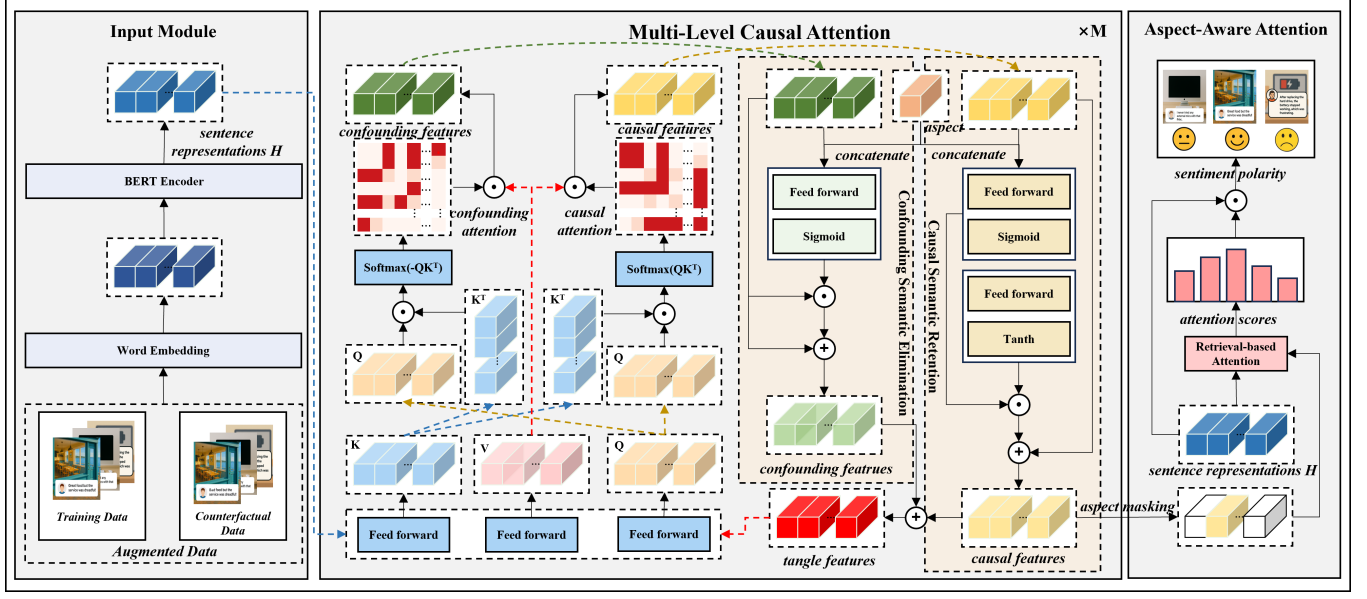


Figure 3: The core component of the overall framework of CECR is the multi-level causal attention module consisting of causal semantic retention and confounding semantic elimination. Causal semantic retention extracts causal features while confounding semantic elimination removes confounding features.

sentiment state is determined to be positive and rarely to be negative or neutral. Deep models based on empirical risk minimization [33] learn from spurious correlations $X_C \leftarrow C \rightarrow P$ and causal correlations $X_C \rightarrow P$ indiscriminately during training. Therefore, we can obtain:

$$P(Y|X) = \frac{1}{||X_L \cup X_C||} \sum_{x_i \in (X_L \cup X_C)} P(Y|x_i) \quad (1)$$

Where $||X_L \cup X_C||$ stands for the number of words in training data X and $Y \in \{y_n, y_o, y_p\}$. Because of the effects of the aforementioned bias factors, $P(y_p|x_i)$ is greater than $P(y_n|x_i)$ and $P(y_o|x_i)$, resulting in $P(Y|x_i)$ is capable of dominating the overall distribution. Therefore, deep models tend to learn $P(Y|X)$ from X_L rather than from X_C during the training process.

Causal intervention. To mitigate spurious correlations, as shown in Fig.2(b), causal intervention is leveraged to guide counterfactual data generation. Specifically, causal words identification, i.e., separating X_L from X_C , is required before implementing causal intervention $do(X_L)$. Afterwards, X_C is varied to change the sentiment polarity P while keeping X_L constant, e.g., $do(X_L) = x_l$. Therefore, after executing the causal intervention on all training data, counterfactual data is utilized to ensure X_C barely contributes to $P(Y|X)$. The rewritten $P(Y|X)$ is computed as follows:

$$P(Y|X) = \frac{1}{||X_C||} \sum_{x_i \in X_C} P(Y|x_i) \quad (2)$$

It means that the model is capable of learning from aspect-related causal terms X_C of $P(Y|X)$.

Counterfactual data generation. Based on causal intervention, as shown in Fig.2 (C), the counterfactual data augmentation module

is composed of three parts: causal term identification, modification, and substitution.

Step1: causal term identification. Identifying causal terms from the training data that are capable of determining the sentiment polarity of the specified aspect.

Step2: causal term modification. Modifying the causal terms while keeping the contextual words unchanged to alter the sentiment polarity of the specified aspect.

Step3: causal term substitution. Generating counterfactual data $\bar{X} = \{(\bar{x}_i, \bar{y}_i)\}_{i=1}^N$ by substituting the identified causal terms with the modified causal terms, where $\bar{y}_i$ has opposite sentiment polarity to y_i .

Since we aim to generate counterfactual data for enhancing the robustness of CECR, the step1, step2, and step3 stages are served with text-completion task prompts. Subsequently, the counterfactual data generated from step3 stage is combined with the training data to form augmented data for training the CECR model to execute the ABSA task.

Prompt design. A triple of prompts $[T_a, T_o, T_x]$ is constructed for each training sample X to guide the LLMs in identifying, modifying, and substituting causal terms to generate the expected counterfactual data. As shown in Fig.2 (C), T_a denotes the step-by-step description of how to generate the counterfactual data in alignment with the ABSA task. T_o denotes the assignment of the output formats. T_x denotes the normalized input format of the training samples x_i .

3.3 BERT Encoder

For input, the counterfactual data $\bar{X}$ is combined with the training samples X to form the augmented data $D = \{(x_i, y_i), (\bar{x}_i, \bar{y}_i)\}_{i=1}^N$ formatted with the sequence $S = \{[CLS] + x_i/\bar{x}_i + [SEP] + a + [SEP]\}$ as input to the text encoder BERT, where $[CLS]$ and $[SEP]$ denote specific tokens:

$$H = \text{BERT}(S) \quad (3)$$

Where $H = \{h_0, \dots, h_{n+m+3}\}$ denote sentence representations of the last layer in BERT.

3.4 Multi-level Causal Attention

We build the multi-level causal attention (MCA) to detect spurious correlations from augmented data D for training debiased models. As illustrated in Fig. 3, MCA employs standard self-attention to compute the key, query, and value vectors of the input features H . After that, the $\text{softmax}(\cdot)$ function is utilized to compute the complementary attentions. The overall function for MCA is presented as follows:

$$\text{MCA} : \begin{cases} q, k, v = W_q H, W_k H, W_v H. \\ c = \text{softmax}(\frac{qk^T}{\sqrt{d_k}})v. \\ s = \text{softmax}(-\frac{qk^T}{\sqrt{d_k}})v_t. \end{cases} \quad (4)$$

Where W_q , W_k , and W_v are three trainable parameters, and d_k denotes the dimension of sentence representations H .

To capture causal features and discard confounding features, the input features x^j (i.e., the output of MCA module at the j th layer) is deconvoluted by the MCA at the $j+1$ th layer into causal features $\hat{c}^{j+1}$ and confounding features $\hat{s}^{j+1}$:

$$\hat{c}^{j+1}, \hat{s}^{j+1} = \text{MCA}(x^j) \quad (5)$$

Then, the confounding features $\hat{s}^{j+1}$ are fed into confounding semantic elimination (CSE) with residual connection by computing the forgetting weights f_s to discard the confounding features:

$$\text{CSE} : \begin{cases} f_s = \sigma(W_{f_s} \cdot [\hat{s}^{j+1}; h_a] + b_{f_s}) \\ s^{j+1} = f_s \cdot \hat{s}^{j+1} + \hat{s}^{j+1} \end{cases} \quad (6)$$

Where W_{f_s} and b_{f_s} are two trainable parameters, h_a denotes the average aspect representations, σ denotes the sigmoid function, and $\cdot$ denotes the concentration operation.

After that, the causal features $\hat{c}^{j+1}$ are fed into the causal semantic retention (CSR) with residual connection to capture causal features c^{j+1} by realizing the interaction of the context with the specified aspect:

$$\text{CSR} : \begin{cases} f_c = \sigma(W_{f_c} \cdot [\hat{c}^{j+1}; h_a] + b_{f_c}) \\ u^{j+1} = \tanh(W_c \cdot [\hat{c}^{j+1}; h_a] + b_c) \\ c^{j+1} = f_c \cdot u^{j+1} + \hat{c}^{j+1} \end{cases} \quad (7)$$

Where W_{f_c} , W_c , b_{f_c} , and b_c are four trainable parameters and $\tanh(\cdot)$ denotes the tanh activation function.

Finally, the CSE chooses to forget the confounding features and the CSR adds candidate causal features to compose the tangle features x^{j+1} :

$$x^{j+1} = c^{j+1} + s^{j+1} \quad (8)$$

Iterating Eq.5-Eq.8 produces a multi-layer MCA. We utilize the final causal features c^M for prediction in our inference.

Table 1: The statistical information of the utilized datasets.

Datasets	Positive		Negative		Neutral	
	Train	Test	Train	Test	Train	Test
Lap14	987	341	866	128	460	169
Rest14	2614	728	805	196	633	196
Twitter	1561	173	1560	173	3127	346
Rest15	1180	440	380	327	50	35
Rest16	1617	590	708	192	88	38
Lap14-ARTS	-	1953	-	473	-	1104
Rest14-ARTS	-	883	-	407	-	587

3.5 Aspect-aware Attention

To emphasize the key features in causal features c^M , the non-aspect representations are concealed by an aspect masking operation, while keeping the aspect representations unaltered:

$$\tilde{c}_i^M = \begin{cases} c_i^M, & \text{if } j \leq i \leq m; \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Therefore, the optimal aspect-related causal representations are obtained by the aspect masking operation as follows:

$$C_{mask} = \{0, \dots, \tilde{c}_j^M, \dots, \tilde{c}_m^M, \dots, 0\}. \quad (10)$$

To capture the most relevant information from contextual representations, the retrieval-based attention is employed to assign attention scores for contextual representations [48]:

$$\beta_t = \sum_{i=0}^n h_i^T \tilde{a}_i = \sum_{i=j}^m h_i^T \tilde{c}_i^M. \quad (11)$$

$$\alpha_t = \frac{\exp(\beta_t)}{\sum_{i=1}^n \exp(\beta_i)}. \quad (12)$$

Here, this paper mainly adopts the retrieval-based attention mechanism to measure the sentiment relevance of optimal aspect-related causal representations and contextual representations. Meanwhile, the aspect masking operation is adopted to further highlight the importance of aspect-related causal representations. Therefore, the final sentiment classification is computed as follows:

$$c_f = \sum_{i=1}^n \alpha_i h_i. \quad (13)$$

$$y = \text{softmax}(W_f c_f + b_f). \quad (14)$$

Where W_f and b_f are two learnable parameters and $\text{softmax}(\cdot)$ denotes the softmax function.

3.6 Loss and Training

The optimization of the objective loss for CECR is presented as follows:

$$\mathcal{L} = \sum_{i=1}^S \sum_{j=1}^C \hat{y}_i^j \log y_i^j + \lambda \|\theta\|_2 \quad (15)$$

Where λ_3 denotes the regularization coefficient. $\|\theta\|_2$ denotes the L_2 regularization trainable parameter θ , S is the number of training samples, C denotes the category labels of the training samples, $\hat{y}$ is the distribution of the correct samples.

Table 2: Results of CECR on utilized datasets compared to data augmented models and deep models. The symbol “-” indicates that the results are not reported in the original paper. The symbol $\dagger$ indicates that the results are derived from the original article and the symbol $\ddagger$ indicates that the results are derived from the [34], [38], and [3]. Bold indicates the optimal results.

Models	Rest14		Lap14		Rest15		Rest16		Twitter	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
Data augmented models										
BT[9] $\ddagger$	86.47	80.29	79.67	76.35	-	-	-	-	76.63	75.47
SCon[16] $\ddagger$	86.51	80.49	80.23	76.48	-	-	-	-	-	-
CBERT[41] $\ddagger$	86.27	80.00	79.83	76.12	-	-	-	-	76.44	75.36
C ³ DA[34] $\dagger$	86.93	81.23	80.61	77.11	-	-	-	-	77.55	76.53
RSC[38] $\dagger$	84.29	76.07	77.74	73.11	82.03	65.80	89.96	73.06	-	-
Deep models										
CapsNet [13] $\dagger$	85.36	78.41	78.97	75.66	82.10	65.57	90.10	75.15	-	-
BERT-SPC [26] $\dagger$	84.11	76.68	77.59	73.28	83.48	66.18	90.10	74.16	-	-
SK-GCN [52] $\dagger$	83.48	75.19	79.00	75.57	83.20	66.78	87.19	72.02	75.00	73.01
R-GAT[37] $\ddagger$	86.60	81.35	78.21	74.07	83.22	69.73	89.71	76.62	76.15	74.48
ACLT [54] $\dagger$	85.71	78.44	79.68	75.83	82.44	72.08	92.15	78.64	75.48	74.51
AG-VSR[11] $\dagger$	86.34	80.88	79.92	75.85	-	-	-	-	76.45	75.04
IAGCN [25] $\dagger$	85.15	75.96	80.42	75.93	84.12	69.25	91.40	74.04	76.64	73.76
ASHGAT[20] $\dagger$	85.49	79.23	79.98	76.58	83.57	71.15	90.75	77.57	-	-
Dual GCN [15] $\dagger$	87.13	81.16	81.80	78.10	81.73	65.05	89.29	68.08	77.40	76.02
TextGT[46] $\dagger$	87.31	82.27	81.33	78.71	-	-	92.21	81.48	77.70	76.45
SVCCCL[39] $\dagger$	86.34	80.16	81.35	77.76	85.24	71.71	91.72	79.07	74.28	72.26
CECR	87.91	82.49	82.84	79.77	87.19	73.26	93.37	81.38	77.98	76.69
CECR w/o CDA	86.19	79.54	82.13	78.67	85.24	71.18	91.68	78.82	75.34	74.27
CECR w/o MCA	87.65	81.74	81.45	77.31	85.98	72.46	92.07	80.50	77.17	75.29

4 Experiments

4.1 Datasets

We examine the performance of CECR on five benchmark datasets: Lap14 and Rest14 derived from SemEval 2014 Task 4 [22], Rest15 derived from SemEval 2015 Task [21], Rest16 derived from SemEval 2016 Task [14], and Twitter derived from ACL14 Task [8]. Moreover, to validate the robustness of CECR, we conduct experiments on ARTS datasets [43], including Rest14-ARTS and Lap14-ARTS. Each sample in the utilized datasets consists of comment sentences, the specified aspect, and the sentiment polarity of the specified aspect (i.e., positive, negative, and neutral). Detailed statistics for the utilized datasets are presented in Table 1.

4.2 Implementation Details

We employ DeepSeek-R1 for identification, modification, and substitution of causal terms in training data. DeepSeek-R1 provides an inference engine with state-of-the-art performance and an innovative architecture that achieves a breakthrough balance between accuracy and efficiency [6]. BERT-base-uncased version is employed as a text encoder, which contains 12 stacked encoders, 12 attention heads, and 768 dimensioned hidden state representations. Moreover,

we employ dropout with a neuron technique that ignores random selection to mitigate oversmoothing during training. The dropout rate of the CSR and CSE modules is set to 0.1. We employ AdamW [32] with the learning rate of $2e-5$ as the optimizer. The number of layers in the MCA module is set to 2. The regularization coefficient λ is set to 10^{-4} . Early stopping is used to stop the training of the model when the accuracy does not improve for five consecutive rounds. To evaluate the performance of the ABSA models, we adopt two widely-used metrics: Accuracy (ACC) and macro-F1 (F1).

4.3 Baseline Methods

To verify the performance of CECR, we make comparisons with state-of-the-art baselines on five benchmark datasets:

Data augmented models: BT [9] translates the original corpus into another language and back-translates it to the original language. CBERT [41] uses token-level substitution to leverage masked language modeling. C³DA [34] devises cross-channel data augmentation. SCon [16] that exploits a novel aspect-invariant data augmentation and contrast-supervised learning framework to recognize valuable sentiment features. RSC [38] that exploits a prompting template to generate appropriate explanations for reducing spurious correlations.

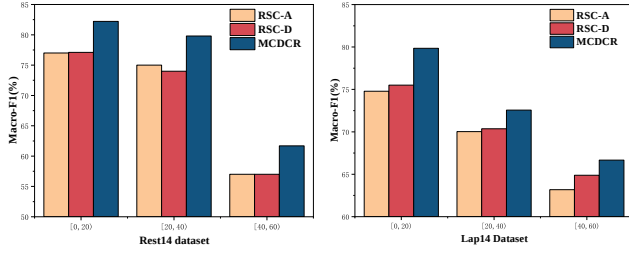


Figure 4: Comparison of CECR with other baselines on samples of different lengths. Note that [20,40) indicates that the length of the sample is greater than or equal to 20 and less than 40, and that [20,40) and [40,60) have the same meaning.

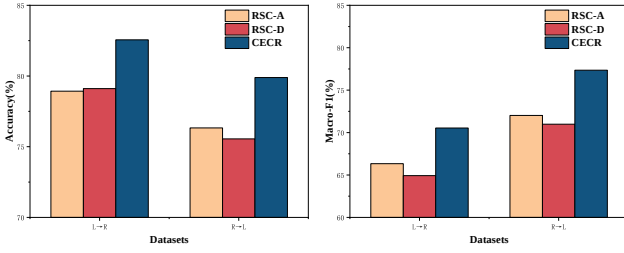


Figure 5: Results of the generalization performance of CECR compared to other baselines. Note that L→R (or R→L) indicates training on the Lap14 (or Rest14) training set, and validated on the Rest14 (or Lap14) test set.

Deep models: CapsNet [13] adopts capsule networks to capture semantic interactions between aspects and contexts. BERT-SPC [26] inputs sentence-aspect pairs to BERT for aspect-aware representations. SK-GCN [52] integrates syntactic dependency trees and common-sense knowledge graphs via GCN to enhance aspect-specific sentence representations. R-GAT [37] designs relationship graphs to encode syntax-related dependencies. ACLT [54] proposes aspect-oriented dependency trees for ABSA, linking aspects to opinion words end-to-end. DualGCN [15] develops BiAffine modules to bridge syntactic and semantic features from SynGCN and SemGCN. AG-VSR [11] designs variational sentence representations for global information encoding. IAGCN [25] employs dynamic weighting layers to minimize irrelevant information and extract contextual features for sentiment analysis. ASHGAT [20] designs aspect-specific hypergraph attention for higher-order interword relationships and key information extraction. SVCCL [39] uses self-attention variational autoencoders to capture explicit aspect-context correlations. TextGT [46] proposes two-view text transformers combining graph and sequence-based learning.

4.4 Main Results

Table 2 illustrates the results of the CECR compared to the state-of-the-art baselines. Subsequently, we provide a detailed analysis of CECR compared to the different types of baselines.

Compared with data augmented models. Analyzing Table 2, CECR exhibits significant superiority compared to data augmented models on all over datasets. This demonstrates that compared to

traditional data augmented approaches (e.g., translation, synonym substitution, random insertion, etc.), the CDA module with the COT inference paradigm generates high-quality counterfactual data by recognizing, modifying, and substituting causal terms. Subsequently, we compare CECR with RSC. CECR and RSC share the common goal of solving the problem of spurious correlations between surface patterns and predicted labels, but there exists a fundamental difference between them. RSC compels LLMs to generate appropriate explanations for the specified aspect of an affective state by constructing cueing templates to account for the cause of the sentiment, while CECR focuses on counterfactual data generation. The results in Table 2 indicate that CECR substantially outperforms RSC, especially on the Rest14, Lap14, and Rest15 datasets.

Comparison with deep models. Reviewing Table 2, CECR significantly outperforms common deep models (e.g., ACLT, IAGCN etc.). These baselines employ graph convolutional neural networks (GCNs) to capture syntactic dependencies between words. Moreover, the supplementation of syntactic dependencies with semantic knowledge generated by self-attention further enhances the GCNs’ performance (e.g., Dual GCN, AG-VSR, etc.). Despite the considerable breakthrough achieved by introducing self-attention into GCNs, there still exists spurious correlations between surface features and predicted labels, which limits further improvement in their performance. In contrast, the MCA module is capable of capturing aspect-related semantic expressions by effectively extracting causal features while removing confounding features. Therefore, compared to the aforementioned baseline models, CECR makes a great success. It is worth noting that TextGT builds image-sequence transformers, which is unfair to us, but CECR still outperforms it by an obvious margin.

4.5 Ablation Study

To validate the effectiveness of counterfactual data augmentation and multi-level causal attention for the robust ABSA model, we conducted ablation experiments on five ABSA datasets. Specifically, we employ original training data rather than augmented data for CECR model training (labeled as “CECR w/o CDA”). Moreover, we disregarded the extraction of causal features and the removal of confounding features by removing the multi-level causal attention, and merely adopted the augmented data as inputs to the CECR model (labeled as “CECR w/o MCA”).

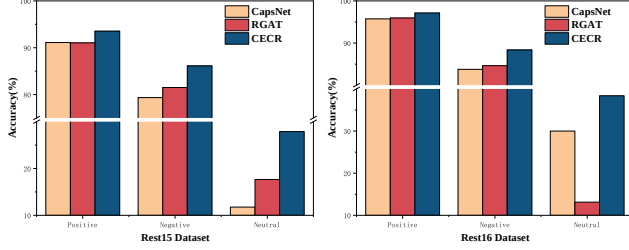
The ablation test results are illustrated in Table 2. When the CDA module is removed, the performance of CECR exhibits a sharp drop. This is because the CDA module generates high-quality counterfactual data, which facilitates the learning of a more robust ABSA model. Furthermore, the contribution of MCA to CECR is particularly significant, which is consistent with our expectation because the MCA module iteratively extracts aspect-related causal features while eliminating the effects of confounding features. It is no surprise that the combination of the CDA and MCA modules provides a significant improvement in CECR.

4.6 Robustness Analysis

To evaluate the robustness of CECR, we conduct experiments on the Aspect Robustness Test Set (ARTS). ARTS verifies the robustness of deep models by constructing three generative strategies,

Table 3: Robustness comparison results of CECR with other baselines on the ARTS test set.

Models	Rest14-ARTS				Lap14-ARTS			
	Acc(Ori→New)	Drop	F1(Ori→New)	Drop	Acc(Ori→New)	Drop	F1(Ori→New)	Drop
BERT-SPC	84.11→57.81	-26.30	76.68→48.08	-28.60	77.59→58.02	-19.57	73.28→54.58	-18.70
CapsNet	85.36→69.24	-16.12	78.41→55.25	-23.16	78.97→60.31	-18.66	75.66→51.50	-24.16
RGAT	86.60→71.64	-14.96	81.35→60.10	-21.25	78.21→66.30	-11.91	74.07→55.68	-18.32
CECR	87.52→80.95	-6.57	82.37→74.28	-8.09	81.74→70.46	-11.28	78.18→67.09	-11.19

**Figure 6: Comparison of the performance of CECR with other baselines on two datasets with long-tailed data distributions. Note that positive indicates the largest sentiment category and neutral indicates the smallest sentiment category.**

i.e., REVTGT (Reversing the original sentiment of a specific aspect), REVNON (Reversing the sentiment of a non-specific aspect), and ADDDIFF (Generating a non-specific aspect of the opposite sentiment of a specified aspect). Therefore, ARTS can detect whether deep models capture aspect-specific relevant opinion words while shielding sentiment information unrelated to the specified aspect. The experimental results are illustrated in Table 3. CECR significantly outperforms the other baselines when injecting adversarial perturbations, particularly on the ARTS-Rest14 dataset.

4.7 Length-wise Performance Analysis

Deep models tend to acquire statistical biases during the process of handling contexts that contain limited information. Therefore, deep models are susceptible to spurious correlations when predicting short texts. Here, we categorize the sample lengths in the lap14 and Rest14 datasets and conduct experiments to reveal the significant advantages of CECR on short text. The experiment results are illustrated in Fig.4. It can be observed that CECR performs better than the other baselines, especially on short texts with sample lengths from 0 to 20. This is because high-quality counterfactual data can motivate CECR to be trained effectively. Meanwhile, the incorporation of extracted causal features empowers CECR to gain a deeper comprehension of the contextual nuances in short texts.

4.8 Generalization Analysis

To examine the generalizability of CECR, we conduct experiments in cross-domain and long-tail scenarios.

For the cross-domain scenario, we trained the proposed CECR model along with other baseline methods on the Lap14 (or Rest14) training dataset and evaluated them on the Rest14 (or Lap14) test

dataset. The experimental results are presented in Fig. 5. It can be observed that CECR consistently outperforms all baseline models, demonstrating its strong robustness in cross-domain settings. Specifically, the proposed CECR framework effectively reduces the spurious correlations between input features and output labels, thereby learning more robust and generalizable feature representations that are closely aligned with the sentiment information of the specified aspect.

For the long-tail scenario, the number of largest classes (positive classes) in the Rest15 and Rest16 datasets in Table 1 is more than 10 times the number of smallest classes (neutral classes). Therefore, we conduct experiments on the aforementioned unbalanced datasets. Fig.6 illustrates the results of the experiments with sentiment categories of positive, neutral, and negative. It can be observed that CECR outperforms the other baseline models, especially on the prediction of the neutral category, verifying that CECR exhibits superior robustness in dealing with long-tailed bias and unbalanced datasets. We attribute this to high-quality factual data augmentation, the capture of aspect-related causal features, and the removal of confounding features to effectively reduce spurious correlations in ABSA.

5 Conclusion

In this paper, we propose a CECR framework to mitigate spurious correlations between input features and output labels. CECR learns a more robust model through CDA and MCA modules. At the data level, inspired by the COT inference paradigm, a three-step-based CDA module is proposed to guide LLMs to perform causal term identification, modification, and substitution for generating high-quality counterfactual data from original training samples. At the model level, the multi-level causal attention (MCA) module comprising causal semantic retention (CSR) and confounding semantic elimination (CSE) facilitates both standard attention capture for aspect-associated causal features and complementary attention extraction for confounding features. These decoupled attentions undergo iterative adversarial minimax optimization, which retains causal attributes while eliminating confounders in an unsupervised paradigm, ultimately enabling debiased model training. Extensive experiments show that CECR outperforms other baselines on five benchmark datasets.

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References

- [1] Murtadha H. M. Ahmed, Qun Chen, and Zhanhuai Li. 2020. Constructing domain-dependent sentiment dictionary for sentiment analysis. *Neural Computing and Applications* 32, 18 (2020), 14719–14732. doi:10.1007/S00521-020-04824-8
- [2] Claire Cardie. 2014. Sentiment Analysis and Opinion Mining Bing Liu (University of Illinois at Chicago) Morgan & Claypool (Synthesis Lectures on Human Language Technologies, edited by Graeme Hirst, 5(1)). 2012, 167 pp; paperbound, ISBN 978-1-60845-884-4. *Comput. Linguistics* 40, 2 (2014), 511–513. doi:10.1162/COLI_R_00186
- [3] Mingshan Chang, Min Yang, Qingshan Jiang, and Ruifeng Xu. 2024. Counterfactual-Enhanced Information Bottleneck for Aspect-Based Sentiment Analysis. In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2024, February 20-27, 2024, Vancouver, Canada*. AAAI Press, 17736–17744. doi:10.1609/AAAI.V38I16.29726
- [4] Peng Chen, Zhongqian Sun, Lidong Bing, and Wei Yang. 2017. Recurrent Attention Network on Memory for Aspect Sentiment Analysis. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing, EMNLP 2017, Copenhagen, Denmark, September 9-11, 2017, Martha Palmer, Rebecca Hwa, and Sebastian Riedel (Eds.)*. Association for Computational Linguistics, 452–461. doi:10.18653/V1/D17-1047
- [5] Christopher Clark, Mark Yatskar, and Luke Zettlemoyer. 2020. Learning to Model and Ignore Dataset Bias with Mixed Capacity Ensembles. In *Findings of the Association for Computational Linguistics: EMNLP 2020, Online Event, 16-20 November 2020 (Findings of ACL, Vol. EMNLP 2020)*, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, 3031–3045. doi:10.18653/V1/2020.FINDINGS-EMNLP.272
- [6] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shiroong Ma, Peiyi Wang, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jiawei Wang, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Meng Li, Miaojuan Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiushi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shutong Pan, and S. S. Li. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *CoRR abs/2501.12948* (2025). arXiv:2501.12948 doi:10.48550/ARXIV.2501.12948
- [7] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2019, Minneapolis, MN, USA, June 2-7, 2019, Volume 1 (Long and Short Papers)*, Jill Burstein, Christy Doran, and Tamar Solorio (Eds.). Association for Computational Linguistics, 4171–4186. doi:10.18653/V1/N19-1423
- [8] Li Dong, Furu Wei, Chuangqi Tan, Duyu Tang, Ming Zhou, and Ke Xu. 2014. Adaptive Recursive Neural Network for Target-dependent Twitter Sentiment Classification. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics, ACL 2014, June 22-27, 2014, Baltimore, MD, USA, Volume 2: Short Papers*. The Association for Computer Linguistics, 49–54. doi:10.3115/V1/P14-2009
- [9] Angela Fan, Shruti Bhosale, Holger Schwenk, Zhiyi Ma, Ahmed El-Kishky, Siddharth Goyal, Mandeep Baines, Onur Celebi, Guillaume Wenzek, Vishrav Chaudhary, Naman Goyal, Tom Birch, Vitaliy Liptchinsky, Sergey Edunov, Michael Auli, and Armand Joulin. 2021. Beyond English-Centric Multilingual Machine Translation. *J. Machine Learning Research* 22 (2021), 107:1–107:48. https://jmlr.org/papers/v22/20-1307.html
- [10] Feifan Fan, Yansong Feng, and Dongyan Zhao. 2018. Multi-grained Attention Network for Aspect-Level Sentiment Classification. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, Brussels, Belgium, October 31 - November 4, 2018, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii (Eds.)*. Association for Computational Linguistics, 3433–3442. doi:10.18653/V1/D18-1380
- [11] Shi Feng, Bing Wang, Zhiyao Yang, and Jihong Ouyang. 2022. Aspect-based sentiment analysis with attention-assisted graph and variational sentence representation. *Knowledge Based System*. 258 (2022), 109975. doi:10.1016/j.knsys.2022.109975
- [12] Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation Artifacts in Natural Language Inference Data. *CoRR abs/1803.02324* (2018). arXiv:1803.02324 http://arxiv.org/abs/1803.02324
- [13] Qingnan Jiang, Lei Chen, Ruifeng Xu, Xiang Ao, and Min Yang. 2019. A Challenge Dataset and Effective Models for Aspect-Based Sentiment Analysis. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019, Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (Eds.)*. Association for Computational Linguistics, 6279–6284. doi:10.18653/V1/D19-1654
- [14] Ayush Kumar, Sarah Kohail, Amit Kumar, Asif Ekbal, and Chris Biemann. 2016. IIT-TUDA at SemEval-2016 Task 5: Beyond Sentiment Lexicon: Combining Domain Dependency and Distributional Semantics Features for Aspect Based Sentiment Analysis. In *Proceedings of the 10th International Workshop on Semantic Evaluation, SemEval@NAACL-HLT 2016, San Diego, CA, USA, June 16-17, 2016*. The Association for Computer Linguistics, 1129–1135. doi:10.18653/V1/S16-1174
- [15] Ruifan Li, Hao Chen, Fangxiang Feng, Zhanyu Ma, Xiaojie Wang, and Eduard H. Hovy. 2024. DualGCN: Exploring Syntactic and Semantic Information for Aspect-Based Sentiment Analysis. *IEEE Transactions on Neural Networks and Learning Systems* 35, 6 (2024), 7642–7656. doi:10.1109/TNNLS.2022.3219615
- [16] Bin Liang, Wangda Luo, Xiang Li, Lin Gui, Min Yang, Xiaoqi Yu, and Ruifeng Xu. 2021. Enhancing Aspect-Based Sentiment Analysis with Supervised Contrastive Learning. In *CIKM '21: The 30th ACM International Conference on Information and Knowledge Management, Virtual Event, Queensland, Australia, November 1-5, 2021*. ACM, 3242–3247. doi:10.1145/3459637.3482096
- [17] R Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. *arXiv preprint arXiv:1902.01007* (2019).
- [18] Sapna Negi and Paul Buitelaar. 2014. INSIGHT Galway: Syntactic and Lexical Features for Aspect Based Sentiment Analysis. In *Proceedings of the 8th International Workshop on Semantic Evaluation, SemEval@COLING 2014, Dublin, Ireland, August 23-24, 2014, Preslav Nakov and Torsten Zesch (Eds.)*. The Association for Computer Linguistics, 346–350. doi:10.3115/V1/S14-2058
- [19] Yixin Nie, Adina Williams, Emily Dinan, Mohit Bansal, Jason Weston, and Douwe Kiela. 2020. Adversarial NLI: A New Benchmark for Natural Language Understanding. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5-10, 2020, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel R. Tetreault (Eds.)*. Association for Computational Linguistics, 4885–4901. doi:10.18653/V1/2020.ACL-MAIN.441
- [20] Jihong Ouyang, Chang Xuan, Bing Wang, and Zhiyao Yang. 2024. Aspect-based sentiment classification with aspect-specific hypergraph attention networks. *Expert Systems with Applications* 248 (2024), 123412. doi:10.1016/j.eswa.2024.123412
- [21] Aitor Garcia Pablos, Montse Cuadros, and German Rigau. 2015. V3: Unsupervised Aspect Based Sentiment Analysis for SemEval2015 Task 12. In *Proceedings of the 9th International Workshop on Semantic Evaluation, SemEval@NAACL-HLT 2015, Denver, Colorado, USA, June 4-5, 2015*. The Association for Computer Linguistics, 714–718. doi:10.18653/V1/S15-2121
- [22] Maria Pontiki, Dimitris Galanis, John Pavlopoulos, Harris Papageorgiou, Ion Androutsopoulos, and Suresh Manandhar. 2014. SemEval-2014 Task 4: Aspect Based Sentiment Analysis. In *Proceedings of the 8th International Workshop on Semantic Evaluation, SemEval@COLING 2014, Dublin, Ireland, August 23-24, 2014*. The Association for Computer Linguistics, 27–35. doi:10.3115/V1/S14-2004
- [23] Jing Ren, Wenhao Zhou, Bowen Li, Mujie Liu, Nguyen Linh Dan Le, Jiade Cen, Liping Chen, Ziqi Xu, Xiwei Xu, and Xiaodong Li. 2025. Causal Prompting for Implicit Sentiment Analysis with Large Language Models. *CoRR abs/2507.00389* (2025). arXiv:2507.00389 doi:10.48550/ARXIV.2507.00389
- [24] Ram Shanmugam. 2001. Causality: Models, Reasoning, and Inference : Judea Pearl; Cambridge University Press, Cambridge, UK, 2000, pp 384, ISBN 0-521-77362-8. *Neurocomputing* 41, 1-4 (2001), 189–190. doi:10.1016/S0925-2312(01)00330-7
- [25] Nikhil Kumar Singh, Sanjay Agal, Thippa Reddy Gadekallu, Mohammad Shabaz, Ismail Keshta, Latika Jindal, Mukesh Soni, Haewon Byeon, and Pavitar Parkash Singh. 2024. Deep Learning Model for Interpretability and Explainability of Aspect-Level Sentiment Analysis Based on Social Media. *IEEE Transactions on Computational Social Systems* (2024).
- [26] Youwei Song, Jiahai Wang, Tao Jiang, Zhiyue Liu, and Yanghui Rao. 2019. Attentional Encoder Network for Targeted Sentiment Classification. *CoRR abs/1902.09314* (2019). arXiv:1902.09314 http://arxiv.org/abs/1902.09314
- [27] Joe Stacey, Pasquale Minervini, Haim Dubossarsky, Sebastian Riedel, and Tim Rocktäschel. 2020. Avoiding the Hypothesis-Only Bias in Natural Language Inference via Ensemble Adversarial Training. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP 2020, Online, November 16-20, 2020, Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu (Eds.)*. Association for Computational Linguistics, 8281–8291. doi:10.18653/V1/2020.EMNLP-MAIN.665
- [28] André Stuhlsatz, Christine Meyer, Florian Eyben, Thomas Zielke, Hans-Günter Meier, and Björn W. Schuller. 2011. Deep neural networks for acoustic emotion recognition: Raising the benchmarks. In *Proceedings of the IEEE International*

- Conference on Acoustics, Speech, and Signal Processing, ICASSP 2011, May 22–27, 2011, Prague Congress Center, Prague, Czech Republic. IEEE, 5688–5691. doi:10.1109/ICASSP.2011.5947651
- [29] Xingwei Tan, Yi Cai, and Changxi Zhu. 2019. Recognizing Conflict Opinions in Aspect-level Sentiment Classification with Dual Attention Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3–7, 2019*, Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (Eds.). Association for Computational Linguistics, 3424–3429. doi:10.18653/V1/D19-1342
- [30] Bing Tian, Yixin Cao, Yong Zhang, and Chunxiao Xing. 2022. Debiasing nlu models via causal intervention and counterfactual reasoning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 11376–11384.
- [31] Bing Tian, Yixin Cao, Yong Zhang, and Chunxiao Xing. 2022. Debiasing NLU Models via Causal Intervention and Counterfactual Reasoning. In *Thirty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2022, Thirty-Fourth Conference on Innovative Applications of Artificial Intelligence, IAAI 2022, The Twelfth Symposium on Educational Advances in Artificial Intelligence, EAAI 2022 Virtual Event, February 22 - March 1, 2022*. AAAI Press, 11376–11384. doi:10.1609/AAAI.V36I10.21389
- [32] Arijit Ukil, Soma Bandyopadhyay, and Arpan Pal. 2020. Sig- R^2 ResNet: Residual Network with Signal Processing-refined Residual Mapping, Auto-tuned L1-Regularization with Modified Adam Optimizer for Time Series Classification. In *2020 International Joint Conference on Neural Networks, IJCNN 2020, Glasgow, United Kingdom, July 19–24, 2020*. IEEE, 1–8. doi:10.1109/IJCNN48605.2020.9206827
- [33] Vladimir Vapnik. 1991. Principles of Risk Minimization for Learning Theory. In *Advances in Neural Information Processing Systems 4, [NIPS Conference, Denver, Colorado, USA, December 2–5, 1991]*, John E. Moody, Stephen Jose Hanson, and Richard Lippmann (Eds.). Morgan Kaufmann, 831–838. <http://papers.nips.cc/paper/506-principles-of-risk-minimization-for-learning-theory>
- [34] Bing Wang, Liang Ding, Qihuang Zhong, Ximing Li, and Dacheng Tao. 2022. A Contrastive Cross-Channel Data Augmentation Framework for Aspect-Based Sentiment Analysis. In *Proceedings of the 29th International Conference on Computational Linguistics, COLING 2022, Gyeongju, Republic of Korea, October 12–17, 2022*. International Committee on Computational Linguistics, 6691–6704. <https://aclanthology.org/2022.coling-1.581>
- [35] Bokun Wang, Yang Yang, Xing Xu, Alan Hanjalic, and Heng Tao Shen. 2017. Adversarial Cross-Modal Retrieval. In *Proceedings of the 2017 ACM on Multimedia Conference, MM 2017, Mountain View, CA, USA, October 23–27, 2017*, Qiong Liu, Rainer Lienhart, Haozhong Wang, Sheng-Wei "Kuan-Ta" Chen, Susanne Boll, Yi-Ping Phoebe Chen, Gerald Friedland, Jia Li, and Shuicheng Yan (Eds.). ACM, 154–162. doi:10.1145/3123266.3123326
- [36] Hanqing Wang, Wei Liang, Jianbing Shen, Luc Van Gool, and Wenguan Wang. 2022. Counterfactual Cycle-Consistent Learning for Instruction Following and Generation in Vision-Language Navigation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18–24, 2022*. IEEE, 15450–15460. doi:10.1109/CVPR52688.2022.01503
- [37] Kai Wang, Weizhou Shen, Yunyi Yang, Xiaojun Quan, and Rui Wang. 2020. Relational Graph Attention Network for Aspect-based Sentiment Analysis. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5–10, 2020*. Association for Computational Linguistics, 3229–3238. doi:10.18653/V1/2020.ACL-MAIN.295
- [38] Qianlong Wang, Keyang Ding, Bin Liang, Min Yang, and Ruifeng Xu. 2023. Reducing Spurious Correlations in Aspect-based Sentiment Analysis with Explanation from Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6–10, 2023*. Association for Computational Linguistics, 2930–2941. doi:10.18653/V1/2023.FINDINGS-EMNLP.193
- [39] Xinyue Wang, Long Liu, Zhuo Chen, Haiyan Wang, and Bin Yu. 2025. Graph convolutional network based on self-attention variational autoencoder and capsule contrastive learning for aspect-based sentiment analysis. *Expert Syst. Appl.* 279 (2025), 127172. doi:10.1016/J.ESWA.2025.127172
- [40] Zhao Wang and Aron Culotta. 2021. Robustness to Spurious Correlations in Text Classification via Automatically Generated Counterfactuals. In *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021, Thirty-Third Conference on Innovative Applications of Artificial Intelligence, IAAI 2021, The Eleventh Symposium on Educational Advances in Artificial Intelligence, EAAI 2021, Virtual Event, February 2–9, 2021*. AAAI Press, 14024–14031. doi:10.1609/AAAI.V35I16.17651
- [41] Xing Wu, Shangwen Lv, Liangjun Zang, Jizhong Han, and Songlin Hu. 2019. Conditional BERT Contextual Augmentation. In *Computational Science - ICCS 2019 - 19th International Conference, Faro, Portugal, June 12–14, 2019, Proceedings, Part IV (Lecture Notes in Computer Science, Vol. 11539)*. Springer, 84–95. doi:10.1007/978-3-030-22747-0_7
- [42] Yuxiang Wu, Matt Gardner, Pontus Stenetorp, and Pradeep Dasigi. 2022. Generating data to mitigate spurious correlations in natural language inference datasets. *arXiv preprint arXiv:2203.12942* (2022).
- [43] Xiaoyu Xing, Zhijing Jin, Di Jin, Bingning Wang, Qi Zhang, and Xuanjing Huang. 2020. Tasty Burgers, Soggy Fries: Probing Aspect Robustness in Aspect-Based Sentiment Analysis. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP 2020, Online, November 16–20, 2020*, Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, 3594–3605. doi:10.18653/V1/2020.EMNLP-MAIN.292
- [44] Ziqi Xu, Debo Cheng, Jiuyong Li, Jixue Liu, Lin Liu, and Kui Yu. 2024. Causal Inference with Conditional Front-Door Adjustment and Identifiable Variational Autoencoder. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7–11, 2024*. OpenReview.net. <https://openreview.net/forum?id=wFF9m4v7oC>
- [45] Tao Yang, Qing Yin, Lei Yang, and Ou Wu. 2022. Aspect-Based Sentiment Analysis with New Target Representation and Dependency Attention. *IEEE Transactions on Affective Computing* 13, 2 (2022), 640–650. doi:10.1109/TAFFC.2019.2945028
- [46] Shuo Yin and Guoqiang Zhong. 2024. TextGT: A Double-View Graph Transformer on Text for Aspect-Based Sentiment Analysis. In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2024, February 20–27, 2024, Vancouver, Canada*. AAAI Press, 19404–19412. doi:10.1609/AAAI.V38I17.29911
- [47] Rowan Zellers, Yonatan Bisk, Roy Schwartz, and Yejin Choi. 2018. SWAG: A Large-Scale Adversarial Dataset for Grounded Commonsense Inference. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, Brussels, Belgium, October 31 - November 4, 2018*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsuiji (Eds.). Association for Computational Linguistics, 93–104. doi:10.18653/V1/D18-1009
- [48] Chen Zhang, Qiuchi Li, and Dawei Song. 2019. Aspect-based Sentiment Classification with Aspect-specific Graph Convolutional Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3–7, 2019*, Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (Eds.). Association for Computational Linguistics, 4567–4577. doi:10.18653/V1/D19-1464
- [49] Wenxuan Zhang, Xin Li, Yang Deng, Lidong Bing, and Wai Lam. 2023. A Survey on Aspect-Based Sentiment Analysis: Tasks, Methods, and Challenges. *IEEE Transactions on Knowledge and Data Engineering* 35, 11 (2023), 11019–11038. doi:10.1109/TKDE.2022.3230975
- [50] Bo Zhao, Yinghao Zhang, Ziqi Xu, Yongli Ren, Xiuzhen Zhang, Renqiang Luo, Zaiwen Feng, and Feng Xia. 2025. Unbiased Reasoning for Knowledge-Intensive Tasks in Large Language Models via Conditional Front-Door Adjustment. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management, CIKM 2025, Seoul, Republic of Korea, November 10–14, 2025*, Meeyoung Cha, Chanyoung Park, Noseong Park, Carl Yang, Senjuti Basu Roy, Jessie Li, Jaap Kamps, Kijung Shin, Bryan Hooi, and Lifang He (Eds.). ACM, 4315–4325. doi:10.1145/3746252.3761103
- [51] Qihuang Zhong, Liang Ding, Juhua Liu, Bo Du, Hua Jin, and Dacheng Tao. 2023. Knowledge Graph Augmented Network Towards Multiview Representation Learning for Aspect-Based Sentiment Analysis. *IEEE Transactions on Knowledge and Data Engineering* 35, 10 (2023), 10098–10111. doi:10.1109/TKDE.2023.3250499
- [52] Jie Zhou, Jimmy Xiangji Huang, Qinmin Vivian Hu, and Liang He. 2020. SK-GCN: Modeling Syntax and Knowledge via Graph Convolutional Network for aspect-level sentiment classification. *Knowledge-Based Systems* 205 (2020), 106292. doi:10.1016/J.KNSYS.2020.106292
- [53] Wujie Zhou, Xun Yang, Xiena Dong, Meixin Fang, Weiqing Yan, and Ting Luo. 2024. MJNet-S*: Multistyle Joint-Perception Network With Knowledge Distillation for Drone RGB-Thermal Crowd Density Estimation in Smart Cities. *IEEE Internet Things J.* 11, 11 (2024), 20327–20339. doi:10.1109/JIOT.2024.3369642
- [54] Yuxiang Zhou, Lejian Liao, Yang Gao, Zhanming Jie, and Wei Lu. 2021. To be Closer: Learning to Link up Aspects with Opinions. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 7–11 November, 2021*. Association for Computational Linguistics, 3899–3909. doi:10.18653/V1/2021.EMNLP-MAIN.317